



José Antonio Viteri Cuenca

ML & DATA ENGINEER

CONTACT

☎ (+49) 17621328747

✉ josevns95@gmail.com

✉ jose.viteri@tum.de

🌐 javiteri95.github.io

📍 Chiemgaustraße 147, 81549, Munich, Germany

PROFILE

Machine Learning & Data Engineer with 8+ years of experience across medical AI, delivery-tech, fintech, and research. Currently pursuing an MSc in Data Engineering & Analytics at TU Munich, with a strong focus on building intelligent, scalable data systems and end-to-end machine learning pipelines.

SKILLS

- **Data Engineering and Analytics:** PyTorch, scikit-learn, XGBoost, LightGBM, Apache Spark, Apache Flink, Apache Kafka, Apache Airflow, LangChain, pandas, NumPy, Matplotlib, Optuna
- **Programming Languages:** Python, C++, C, Java, Go, JavaScript, TypeScript, Bash
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Snowflake, Qdrant
- **Web Development:** FastAPI, Django, Flask, Gin, React, Vue, Express
- **Infrastructure:** Docker, Kubernetes, AWS, Git, Linux, HPC

WORK EXPERIENCE

- **Research Assistant – Helmholtz Zentrum München** 01/11/2024 – Current
Munich, Germany
 - Designed, developed, and maintained a RAG application for internal use at the research center, enabling efficient access to domain-specific knowledge from web sources, internal documents, and knowledge graphs.
 - Designed and implemented a data pipeline to ingest and process data for the RAG application.
 - Designed and implemented an agentic application to help users answer questions in HTML forms.
- **Data Engineer (Intern) – Reply DE** 01/10/2025 – 13/03/2026
Munich, Germany
 - Performed an energy-consumption analysis of major data engineering platforms (Spark, Kafka, Flink).
 - Designed a data lake architecture using the most efficient components to reduce the system's overall energy footprint.
- **Machine Learning Engineer – Rappi** 18/07/2022 – 10/04/2024
Guayaquil, Ecuador
 - Developed and maintained microservices for models that predict ETAs (Estimated Time of Arrival) across the full lifecycle of an order at Rappi.
 - Built and maintained data pipelines to ensure correct online training and validation of those models.
- **Deep Learning Engineer – Tiendas Industriales Asociadas S.A. (TIA S.A.)** 05/12/2020 – 31/05/2022
Guayaquil, Ecuador
 - Developed a Deep Learning (DL) system that generates customer heatmaps from video surveillance camera systems.
 - Developed a DL system that counts the number of people in a retail store using video surveillance camera systems.
 - Designed and developed a medical DL system capable of inferring the presence of both Diabetic Retinopathy and Glaucoma from patients' retinal images.
- **Full Stack Web Developer – Criptext** 25/11/2018 – 02/12/2020
Guayaquil, Ecuador
 - Developed part of the API and the Bob encryption service of Criptext, an encrypted email provider.
 - Built the admin web application for enterprise stakeholders and contributed to the Electron-based desktop application.
 - Handled DevOps tasks, maintaining and operating Criptext servers.
- **University Research Assistant – Escuela Superior Politécnica del Litoral** 04/09/2016 – 22/02/2019
Guayaquil, Ecuador
 - Developed a YCSB-based key-value store benchmark that replays real workloads and traces to analyze performance, latency, and throughput across various key-value databases.
 - Designed a cache replacement and admission policy that automatically tunes partition sizes.
- **Web Developer (Intern) – Procare Foundation** 04/09/2016 – 04/10/2017
Guayaquil, Ecuador
 - Built a REST API powering a desktop app to administer the Procare Foundation's teachers, students, staff, classes, and inventory.
 - Contributed to the development of the desktop application itself.

EDUCATION & TRAINING

INTERESTS

I love to read, and I'm always looking for a new book to dive into. Moreover, going outside, for a walk or a hike, is one of my favorite ways to clear my mind. I also love to travel, exploring new places. And, not less importantly, I'm always up for a good dance or a good beer.

REFERENCES

Marie Piraud

Head of the Helmholtz AI Consultant Team
· Helmholtz Munich
marie.piraud@helmholtz-munich.de
Ingolstädter Landstraße 1, 85764 Neuherberg, Germany

Karol Szustakowski

AI Consultant · Helmholtz Munich
karol.szustakowski@helmholtz-munich.de
Ingolstädter Landstraße 1, 85764 Neuherberg, Germany

Antonio Di Turi

Manager · Reply DE
a.dituri@reply.de
Luise-Ullrich-Straße 14, 80636 Munich, Germany

Cornelius Wolf

PhD student · CWI (Centrum Wiskunde & Informatica)
Cornelius.Wolff@cwi.nl
Science Park 123, 1098 XG Amsterdam, Netherlands

- **Master of Science in Data Engineering and Analytics** 01/04/2024 – Current
Technical University of Munich · Munich, Germany
Website: <https://www.tum.de/en/studies/degree-programs/detail/data-engineering-and-analytics-master-of-science-msc>
- **Bachelor of Science in Computer Science** 06/05/2014 – 22/02/2020
Escuela Superior Politécnica del Litoral (ESPOL) · Guayaquil, Ecuador
Website: <https://www.espol.edu.ec/es>

CERTIFICATIONS

- **Machine Learning in Python with Scikit-Learn (MOOC)** 01/05/2022 – 17/05/2022
France Université Numérique
Website: <https://www.fun-mooc.fr/en/courses/machine-learning-python-scikit-learn/>
- **C++ for C Programmers** 28/07/2020 – 04/08/2020
University of California Santa Cruz
Website: <https://www.coursera.org/learn/c-plus-plus-a>

FREELANCING & PROJECTS

- **Master's Thesis Researcher – CWI & Technical University of Munich** 03/2026 – Current
Munich, Germany / Amsterdam, Netherlands (Remote)
 - Investigating structural and semantic ambiguity mitigation workflows within Text-to-SQL translation systems using Large Language Models.
 - Built a dialogue system that utilizes Expected Information Gain (EIG) metrics to dynamically generate user feedback prompts, minimizing total conversational steps while maximizing retrieval accuracy.
- **Data Engineer – Technical University of Munich** 15/10/2025 – 30/05/2026
Munich, Germany
 - Conducted exploratory data analysis on Reddit posts and a literature review on long COVID.
 - Built a knowledge graph from those Reddit posts and literature sources on long COVID.
- **Data Scientist – Escuela Superior Politécnica del Litoral** 01/06/2022 – 31/12/2023
Guayaquil, Ecuador
 - Carried out a trace analysis of Microsoft Azure Functions.
 - Performed benchmarking of tools such as Apache Spark, Presto, and Amazon Athena.
- **Data Scientist – Escuela Superior Politécnica del Litoral** 05/09/2021 – 12/12/2021
Guayaquil, Ecuador
 - Performed exploratory data analysis on Microsoft's Azure Functions trace dataset.
 - Carried out time-series analysis (trends, burstiness, and seasonality) and designed policies to reduce burstiness in the dataset.
- **Data Scientist – Escuela Superior Politécnica del Litoral** 05/11/2019 – 21/02/2020
Guayaquil, Ecuador
 - Developed a Convolutional Neural Network (CNN) to automatically detect and segment White Matter Hyperintensities in the brain, achieving state-of-the-art results on the topic.

- Built a desktop application to visualize the brain and its lesions, branded under the name Brainy.

• Web Developer – LACEA LAMES

02/04/2018 – 10/11/2018

Guayaquil, Ecuador

- Built a hybrid mobile app for the LACEA LAMES 2018 Econometrics and Economics conference, providing schedules, bus routes, tourist attractions, and session notifications.
- Developed an admin web application so organizers could manage the different conference matters.
- Implemented a REST API that fed both the web and mobile applications.

• Software Developer – Escuela Superior Politécnica del Litoral

01/10/2017 – 18/02/2018

Guayaquil, Ecuador

- Built a native Android application to monitor the Bosque Protector Prosperina at ESPOL.
- Implemented a data-streaming service for communication between the app and a remote server.
- Developed an audio-listening service that registered events based on the smartphone's location.

LANGUAGES

Spanish: Mother tongue | English: C1 | German: B1

PUBLICATIONS

Google Scholar | ORCID: 0000-0002-9115-6881

• Automatic Brain White Matter Hyperintensities Segmentation with Swin U-Net

2022

José A Viteri, Bryan V Piguave, Enrique Pelaez, Francis R Loayza, 2022 IEEE ANDESCON, 1-6

Proposes a Swin Transformer-based U-Net for automatic segmentation of White Matter Hyperintensities in brain MRI, using hierarchical self-attention to improve lesion delineation over purely convolutional architectures.

• Automatic Brain White Matter Hyperintensities Segmentation using Deep Learning Techniques

2021

José A Viteri, Francis R Loayza, Enrique Pelaez, Fabricio Layedra, 2021 HEALTHINF, 244-252

A 21-layer convolutional U-Net for automatic WMH segmentation from FLAIR and T1 MRI, trained on the MICCAI WMH Challenge dataset; ranked 9th of 43 teams, with its strongest performance on the Hausdorff-distance metric.

• Benchmarking Key-Value Stores via Trace Replay

2017

Edwin F Boza, Cesar San-Lucas, Cristina L Abad, Jose A Viteri, 2017 IC2E, 183-189

Introduces a trace-replay benchmarking methodology that drives key-value stores with real captured workloads, enabling apples-to-apples comparison of latency and throughput across different database systems.

HONOURS & AWARDS

• Best graduate of the bachelor's degree in computer science

31/12/2020

Escuela Superior Politécnica del Litoral (ESPOL)

Selected as the best graduate of the bachelor's degree in computer science for the winter semester 2019/2020. [Link](#)

• Best undergraduate “5-minute pitch” project

31/12/2020

Facultad de Ingeniería en Electricidad y Computación (FIEC)

Selected as the best undergraduate “5-minute pitch” project of Facultad de Ingeniería en Electricidad y Computación (FIEC).

- **First place in ESPOL and second place in Ecuador at the IEEEExtreme 24-hour programming competition** 22/10/2016

IEEE

Programming contest organized by IEEE that involved solving medium- and hard-level programming problems. First place in ESPOL. Second place in Ecuador.

NETWORKS & MEMBERSHIPS

- **TuMuchData** 05/2026 – Current

Munich

Student club specialized in Database Systems and Data Engineering.

- **TAWS** 04/2016 – 05/2020

Guayaquil

Student club specialized in Data Science and Machine Learning.

- **Argumentum** 09/2014 – 05/2020

Guayaquil

Student club specialized in enhancing social and oral skills, United Nations simulations, and competitive debate.